Xi'an Jiaotong-Liverpool University 西交利物浦大学

MODULE CODE	EXAMINER	DEPARTMENT	TEL
ACC102	Xiangyun Lu	IBSS	6530

2nd SEMESTER 2024-25

Mini Assignment: Python Data Product (15%)

BA Accounting – Year 2

BSc Economics

BSc Economics and Finance

Final

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1. Assignment Objective

This is an individual assignment. You are required to design and publish one small data product using Python and a business, financial, accounting, economics, marketing, operations, or closely related dataset.

The assignment assesses your foundational ability to use Python to obtain, clean, transform, analyse, and communicate data for a clearly defined analytical purpose.

The task is intended to support authentic assessment, portfolio development, and product thinking. Your submission should show a coherent workflow from problem definition to Python-based analysis, insight generation, and a user-facing output.

2. Assignment Overview

You must choose one approved track and build one small data product for a clearly defined user or audience.

All tracks must include a substantive Python component. A polished presentation cannot compensate for weak or minimal analytical work.

AI tools may be used where appropriate, but Python-based analysis remains the core requirement.

You should approach the task from a product or media perspective rather than as a conventional essay.

3. Approved Tracks

- Track 1 – Social Media Data Story: Social Media Data Story. Create and publish a 3–5 minute video

explaining insights derived from your Python analysis. Submit the video link, Python notebook, and reflection report.

- Track 2 – GitHub Data Analysis Project. Publish a Python-based data analysis project on GitHub. Submit the repository link, Python notebook or code files, README, a 1–3 minute demo video, and reflection report. This track may receive up to 10 bonus marks, capped at 100 overall.
- Track 3 – Coze/Dify Data Analysis Agent. Publish a Coze/Dify agent supported by a substantive Python workflow. Submit the Coze/Dify link, Python notebook, workflow evidence, a 1–3 minute demo video, and reflection report. This track may receive up to 10 bonus marks, capped at 100 overall.
- Track 4 – Interactive Data Analysis Tool. Publish a small interactive Python tool, such as a Streamlit app or interactive notebook, on GitHub or another accessible platform. Submit the product link, code files, README, a 1–3 minute demo video, and reflection report. This track may receive up to 2s0 bonus marks, capped at 100 overall.

4. Common Requirements for All Tracks

- Use a dataset that is relevant to a business-related context.
- Use Python to perform substantive data work such as data acquisition, cleaning, transformation, descriptive analysis, visualisation, or simple modelling appropriate to this module level.
- Define a clear analytical problem and a clear target user or audience.
- Produce a final product that communicates analytical value rather than only displaying raw code.
- Demonstrate your own understanding of the methods used.
- Simple but well-executed analysis is preferable to an over-ambitious product that you do not understand.

5. Data Source Guidance

- Suitable sources may include WRDS, Yahoo Finance, World Bank, IMF, government open-data portals, Kaggle, company reports, or responsibly collected web data.
- You are responsible for ensuring that the data are lawful to use, sufficiently reliable for educational analysis, and appropriately acknowledged.
- In your notebook or reflection, you should clearly state the source of the data and the date on which the data were accessed or collected.

6. Submission Requirements

- You must submit all required components via Learning Mall Core (LMO) by **23:59 on 27 April 2026**.
- Your submission must include: (a) one product link appropriate to the chosen track; (b) one Python notebook showing the analytical workflow; and (c) one reflection report of 500–800 words.
- Track-specific extras: Track 1 must meet the 3–5 minute video requirement. Tracks 2 and 4 must include a README. Tracks 2, 3, and 4 must include a 1–3 minute demo video.
- External links do not have to be fully public, but they must be accessible to the marker via the submitted link during the marking period.
- External links alone are not sufficient. The required notebook and reflection report must also be submitted through LMO.
- If your work depends on additional files, packages, or setup steps, these must be clearly explained in the notebook or README.
- All submitted materials, including notebooks, reflection reports, README files, and video narration or subtitles, must be in English.
- You are not required to use your own voice in video submissions. Text-to-speech tools are acceptable. If you need a text-to-speech service, MiniMax is one option you may consider.

7. Python Notebook Requirements

The University attaches penalties to the late submission of assessed coursework. You need to be familiar with the rules as detailed in the University's 'Code of Practice on Assessment'.

8. Python Notebook Requirements

- The notebook must show a coherent analytical workflow from data input to final output.
- It should normally include problem definition, data loading, data cleaning or preparation, analysis, and outputs such as tables, charts, or other product-relevant results.
- Use markdown cells, comments, and clear structure so that the marker can follow your reasoning.
- You do not need to build a highly advanced model, but the Python work must be substantive and relevant to your chosen product.

9. Reflection Report Requirements (500–800 words)

Your reflection report must address all of the following:

- State the analytical problem and intended user or audience.
- Describe the dataset and explain why it was selected.

- Explain the Python methods used.
- Summarise the main insights or outputs produced.
- Evaluate limitations, reliability issues, and possible improvements.
- Briefly explain your own contribution, decisions, and what you learned from the task.

10. AI Use and Academic Integrity

- AI tools may be used where appropriate to support your workflow.
- You must disclose any AI use. This disclosure should be included at the end of the reflection report, following the AI disclosure requirement in week 1 open question.
- For each AI tool used, state the tool name, model or version where available, access date, and what the AI was used for.
- Your submission must still demonstrate your own intellectual contribution and understanding. Undisclosed or **excessive reliance** on AI-generated work may constitute an academic offence under the University Academic Integrity Policy.

11. Marking and Bonus Marks

- The mini assignment is marked out of 100 and contributes 15% to the overall module mark.
- The University grading scale applies: Fail 0–39, Adequate/Basic 40–49, Satisfactory/Competent 50–59, Good 60–69, Very Good 70–79, Excellent to Outstanding 80+.
- Bonus marks reward additional technical or product complexity in Tracks 2, 3, and 4. Bonus marks do not replace the need to meet the core assignment requirements and the final recorded mark cannot exceed 100.

12. Evaluation Criteria

The marking scheme below is designed to support parity across assessors and to make expectations transparent to students.

Criteria	Marks	What this means
Problem Definition and Data Relevance	20	Clarity of the analytical problem, suitability of the dataset, and relevance to a business-related context.
Python Implementation and Technical Execution	30	Correctness, completeness, and appropriateness of the Python workflow.

Analysis, Insight, and Interpretation	20	Quality and relevance of findings, and whether the analysis goes beyond surface description.
Product Design, Communication, and User Focus	20	How well the final product communicates value to the intended user or audience.
Reflection, Limitations, and Professional Practice	10	Quality of reflection, acknowledgement of limitations, and responsible practice including AI disclosure where relevant.

Appendix A

Assessment Form for Mini Assignment (ACC102)

Student ID	
Student Name	
Track	
Product Title	
Marker	

Criteria	Max Point	Mark
Problem Definition and Data Relevance	20	
Python Implementation and Technical Execution	30	
Analysis, Insight, and Interpretation	20	
Product Design, Communication, and User Focus	20	
Reflection, Limitations, and Professional Practice	10	
Total Mark	100	

Comments:

Name and Signature of Examiner (Date):